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ON PERMUTATIONS OF $\{1, \dots, n\}$ AND RELATED TOPICS

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ABSTRACT. In this paper we study combinatorial aspects of permutations of $\{1, \dots, n\}$ and related topics. In particular, we prove that there is a unique permutation π of $\{1, \dots, n\}$ such that all the numbers $k + \pi(k)$ ($k = 1, \dots, n$) are powers of two. We also show that $n \mid \text{per}[i^{j-1}]_{1 \leq i, j \leq n}$ for any integer $n > 2$. We conjecture that if a group G contains no element of order among $2, \dots, n+1$ then any $A \subseteq G$ with $|A| = n$ can be written as $\{a_1, \dots, a_n\}$ with $a_1, a_2^2, \dots, a_n^n$ pairwise distinct. This conjecture is confirmed when G is a torsion-free abelian group. We also prove that for any finite subset A of a torsion-free abelian group G with $|A| = n > 3$, there is a numbering $a_1, \dots, a_n$ of all the elements of A such that all the n sums

$$a_1 + a_2 + a_3, a_2 + a_3 + a_4, \dots, a_{n-2} + a_{n-1} + a_n, a_{n-1} + a_n + a_1, a_n + a_1 + a_2$$

are pairwise distinct, and conjecture that this remains valid if G is cyclic.

1. INTRODUCTION

As usual, for $n \in \mathbb{Z}^+ = \{1, 2, 3, \dots\}$ we let S_n denote the symmetric group of all the permutation of $\{1, \dots, n\}$.

Let $A = [a_{ij}]_{1 \leq i, j \leq n}$ be a $(0, 1)$ -matrix (i.e., $a_{ij} \in \{0, 1\}$ for all $i, j = 1, \dots, n$). Then the permanent of A given by

$$\text{per}(A) = \sum_{\pi \in S_n} a_{1\pi(1)} \cdots a_{n\pi(n)}$$

is just the number of permutations $\pi \in S_n$ with $a_{k\pi(k)} = 1$ for all $k = 1, \dots, n$.

In 2002, B. Cloitre proposed the sequence [Cl, A073364] on OEIS whose n -th term $a(n)$ is the number of permutations $\pi \in S_n$ with $k + \pi(k)$ prime for all $k = 1, \dots, n$. Clearly, $a(n) = \text{per}(A)$, where A is a matrix of order n whose

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(i, j) -entry ($1 \leq i, j \leq n$) is 1 or 0 according as $i + j$ is prime or not. In 2018 P. Bradley [Br] proved that $a(n) > 0$ for all $n \in \mathbb{Z}^+$.

Our first theorem is an extension of Bradley's result.

Theorem 1.1. *Let $(a_1, a_2, \dots)$ be an integer sequence with $a_1 = 2$ and $a_k < a_{k+1} \leq 2a_k$ for all $k = 1, 2, 3, \dots$. Then, for any positive integer n , there exists a permutation $\pi \in S_n$ with $\pi^2 = I_n$ such that*

$$\{k + \pi(k) : k = 1, \dots, n\} \subseteq \{a_1, a_2, \dots\}, \quad (1.1)$$

where I_n is the identity of S_n with $I_n(k) = k$ for all $k = 1, \dots, n$.

Recall that the Fibonacci numbers $F_0, F_1, \dots$ and the Lucas numbers $L_0, L_1, \dots$ are defined by

$$F_0 = 0, \quad F_1 = 1, \quad \text{and} \quad F_{n+1} = F_n + F_{n-1} \quad (n = 1, 2, 3, \dots),$$

and

$$L_0 = 2, \quad L_1 = 1, \quad \text{and} \quad L_{n+1} = L_n + L_{n-1} \quad (n = 1, 2, 3, \dots).$$

If we apply Theorem 1.1 with the sequence $(a_1, a_2, \dots)$ equal to $(F_3, F_4, \dots)$ or $(L_0, L_2, L_3, \dots)$, then we immediately obtain the following consequence.

Corollary 1.1. *Let $n \in \mathbb{Z}^+$. Then there is a permutation $\sigma \in S_n$ with $\sigma^2 = I_n$ such that all the sums $k + \sigma(k)$ ($k = 1, \dots, n$) are Fibonacci numbers. Also, there is a permutation $\tau \in S_n$ with $\tau^2 = I_n$ such that all the numbers $k + \tau(k)$ ($k = 1, \dots, n$) are Lucas numbers.*

Remark 1.1. Let $f(n)$ be the number of permutations $\sigma \in S_n$ such that all the sums $k + \sigma(k)$ ($k = 1, \dots, n$) are Fibonacci numbers. Via **Mathematica** we find that

$$(f(1), \dots, f(20)) = (1, 1, 1, 2, 1, 2, 4, 2, 1, 4, 4, 20, 4, 5, 1, 20, 24, 8, 96, 200).$$

For example, $\pi = (2, 3)(4, 9)(5, 8)(6, 7)$ is the unique permutation in S_9 such that all the numbers $k + \pi(k)$ ($k = 1, \dots, 9$) are Fibonacci numbers.

Recall that those integers $T_n = n(n+1)/2$ ($n = 0, 1, 2, \dots$) are called triangular numbers. Note that $T_n - T_{n-1} = n \leq T_{n-1}$ for every $n = 3, 4, \dots$. Applying Theorem 1.1 with $(a_1, a_2, a_3, \dots) = (2, T_2, T_3, \dots)$, we immediately get the following corollary.

Corollary 1.2. *For any $n \in \mathbb{Z}^+$, there is a permutation $\pi \in S_n$ with $\pi^2 = I_n$ such that each of the sums $k + \pi(k)$ ($k = 1, \dots, n$) is either 2 or a triangular number.*

Remark 1.2. When $n = 4$, we may take $\pi = (2, 4)$ to meet the requirement in Corollary 1.2. Note that $1 + 1 = 3 = T_2$ and $2 + 4 = 3 + 3 = T_3$.

Our next theorem focuses on permutations involving powers of two.

Theorem 1.2. *Let n be any positive integer. Then there is a unique permutation $\pi_n \in S_n$ such that all the numbers $k + \pi_n(k)$ ($k = 1, \dots, n$) are powers of two. In other words, for the $n \times n$ matrix A whose (i, j) -entry is 1 or 0 according as $i + j$ is a power of two or not, we have $\text{per}(A) = 1$.*

Remark 1.3. Note that the number of 1's in the matrix A given in Theorem 1.2 coincides with

$$\sum_{k=0}^{\lfloor \log_2 n \rfloor + 1} \sum_{\substack{1 \leq i, j \leq n \\ i+j=2^k}} 1 = \sum_{k=0}^{\lfloor \log_2 n \rfloor} (2^k - 1) + \sum_{i=2^{\lfloor \log_2 n \rfloor + 1} - n}^n 1 = 2n - \lfloor \log_2 n \rfloor - 1.$$

Example 1.1. Here we list π_n in Theorem 1.2 for $n = 1, \dots, 11$:

$$\begin{aligned} \pi_1 &= (1), \pi_2 = (1), \pi_3 = (1, 3), \pi_4 = (1, 3), \pi_5 = (3, 5), \pi_6 = (2, 6)(3, 5), \\ \pi_7 &= (1, 7)(2, 6)(3, 5), \pi_8 = (1, 7)(2, 6)(3, 5), \pi_9 = (2, 6)(3, 5)(7, 9), \\ \pi_{10} &= (3, 5)(6, 10)(7, 9), \pi_{11} = (1, 3)(5, 11)(6, 10)(7, 9). \end{aligned}$$

Theorem 1.2 has the following consequence.

Corollary 1.3. *For any $n \in \mathbb{Z}^+$, there is a unique permutation $\pi \in S_{2n}$ such that $k + \pi(k) \in \{2^a - 1 : a \in \mathbb{Z}^+\}$ for all $k = 1, \dots, 2n$.*

Now we turn to our results of new types.

Theorem 1.3. (i) *Let p be any odd prime. Then there is no $\pi \in S_{p-1}$ such that all the $p - 1$ numbers $k\pi(k)$ ($k = 1, \dots, p - 1$) are pairwise incongruent modulo p . Also,*

$$\text{per}[i^{j-1}]_{1 \leq i, j \leq p-1} \equiv 0 \pmod{p}. \quad (1.2)$$

(ii) *We have*

$$\text{per}[i^{j-1}]_{1 \leq i, j \leq n} \equiv 0 \pmod{n} \text{ for all } n = 3, 4, 5, \dots \quad (1.3)$$

Remark 1.4. In contrast with Theorem 1.3, it is well-known that

$$\det[i^{j-1}]_{1 \leq i, j \leq n} = \prod_{1 \leq i < j \leq n} (j - i) = 1!2! \dots (n-1)!$$

and in particular

$$\det[i^{j-1}]_{1 \leq i, j \leq p-1}, \det[i^{j-1}]_{1 \leq i, j \leq p} \not\equiv 0 \pmod{p}$$

for any odd prime p .

In additive combinatorics, there are some interesting topics involving both permutations and finite abelian groups, see. e.g., [FSX] and [GS]. Below we present two novel theorems on permutations involving groups.

Theorem 1.4. (i) Let $a_1, \dots, a_n$ be distinct elements of a torsion-free abelian group G . Then there is a permutation $\pi \in S_n$ such that all those $ka_{\pi(k)}$ ($k = 1, \dots, n$) are pairwise distinct.

(ii) Let a, b, c be three distinct elements of a group G such that none of them has order 2 or 3. Then $a^{\sigma(1)}$ and $b^{\sigma(2)}$ are distinct for some $\sigma \in S_2$. Also, $a^{\tau(1)}, b^{\tau(2)}, c^{\tau(3)}$ are pairwise distinct for some $\tau \in S_3$.

Remark 1.5. On the basis of this theorem, we will formulate a general conjecture for groups in Section 4.

Theorem 1.5. For any $n > 3$ distinct elements $a_1, a_2, \dots, a_n$ of a torsion-free abelian group G , there is a permutation $\pi \in S_n$ such that all the n sums

$$b_1 + b_2 + b_3, b_2 + b_3 + b_4, \dots, b_{n-2} + b_{n-1} + b_n, b_{n-1} + b_n + b_1, b_n + b_1 + b_2$$

are pairwise distinct, where $b_k = a_{\pi(k)}$ for $k = 1, \dots, n$.

Remark 1.6. By Remark 1.2 of Sun [S19], for any finite subset A of a torsion-free abelian group with $|A| = n > 2$ we may write A as $\{a_1, \dots, a_n\}$ such that $a_1 + a_2, \dots, a_{n-1} + a_n, a_n + a_1$ are pairwise distinct.

We are going to prove Theorems 1.1-1.3 and Corollary 1.3 in the next section, and show Theorems 1.4-1.5 in Section 3. We will pose some conjectures in Section 4.

2. PROOFS OF THEOREMS 1.1-1.3 AND COROLLARY 1.3

Proof of Theorem 1.1. For convenience, we set $a_0 = 1$ and $A = \{a_1, a_2, a_3, \dots\}$. We use induction on $n \in \mathbb{Z}^+$ to show the desired result.

For $n = 1$, we take $\pi(1) = 1$ and note that $1 + \pi(1) = 2 = a_1 \in A$.

Now let $n \geq 2$ and assume the desired result for smaller values of n . Choose $k \in \mathbb{N}$ with $a_k \leq n < a_{k+1}$, and write $m = a_{k+1} - n$. Then $1 \leq m \leq 2a_k - n \leq 2n - n = n$. Let $\pi(j) = a_{k+1} - j$ for $j = m, \dots, n$. Then

$$\{\pi(j) : j = m, \dots, n\} = \{m, \dots, n\},$$

and $\pi(\pi(j)) = j$ for all $j = m, \dots, n$.

Case 1. $m = 1$.

In this case, $\pi \in S_n$ and $\pi^2 = I_n$.

Case 2. $m = n$.

In this case, $a_{k+1} = 2n \geq 2a_k$. On the other hand, $a_{k+1} \leq 2a_k$. So, $a_{k+1} = 2a_k$ and $a_k = n$. Let $\pi(j) = n - j = a_k - j$ for all $0 < j < n$. Then $\pi \in S_n$ and $j + \pi(j) \in \{a_k, a_{k+1}\}$ for all $j = 1, \dots, n$. Note that $\pi^2(k) = k$ for all $k = 1, \dots, n$.

Case 3. $1 < m < n$.

In this case, by the induction hypothesis, for some $\sigma \in S_{m-1}$ with $\sigma^2 = I_{m-1}$, we have $i + \sigma(i) \in A$ for all $i = 1, \dots, m-1$. Let $\pi(i) = \sigma(i)$ for all $i = 1, \dots, m-1$. Then $\pi \in S_n$ and it meets our requirement.

In view of the above, we have completed the induction proof. $\square$

Proof of Theorem 1.2. Applying Theorem 1.1 with $a_k = 2^k$ for all $k \in \mathbb{Z}^+$, we see that for some $\pi \in S_n$ with $\pi^2 = I_n$ all the numbers $k + \pi(k)$ ($k = 1, \dots, n$) are powers of two.

Below we use induction on n to prove that the number of $\pi \in S_n$ with

$$\{k + \pi(k) : k = 1, \dots, n\} \subseteq \{2^a : a \in \mathbb{Z}^+\}$$

is exactly one.

The case $n = 1$ is trivial.

Now let $n > 1$ and assume that for each $m = 1, \dots, n-1$ there is a unique $\pi_m \in S_m$ such that all the numbers $k + \pi_m(k)$ ($k = 1, \dots, m$) are powers of two. Choose $a \in \mathbb{Z}^+$ with $2^{a-1} \leq n < 2^a$, and write $m = 2^a - n$. Then $1 \leq m \leq n$.

Suppose that $\pi \in S_n$ and all the numbers $k + \pi(k)$ ($k = 1, \dots, n$) are powers of two. If $2^{a-1} \leq k \leq n$, then

$$2^{a-1} < k + \pi(k) \leq k + n \leq 2n < 2^{a+1}$$

and hence $\pi(k) = 2^a - k$ since $k + \pi(k)$ is a power of two. Thus

$$\{\pi(k) : k = 2^{a-1}, \dots, n\} = \{m, \dots, 2^{a-1}\}.$$

If $k \in \{1, \dots, 2^{a-1} - 1\}$ and $2^{a-1} < \pi(k) \leq n$, then

$$2^{a-1} < k + \pi(k) \leq n + n < 2^{a+1},$$

hence $k + \pi(k) = 2^a = m + n$ and thus $m \leq k < 2^{a-1}$. So we have

$$\{\pi^{-1}(j) : 2^{a-1} < j \leq n\} = \{m, \dots, 2^{a-1} - 1\}.$$

(Note that $n - 2^{a-1} = 2^a - m - 2^{a-1} = 2^{a-1} - m$.)

By the above analysis, $\pi(k) = 2^a - k$ for all $k = m, \dots, n$, and

$$\{\pi(k) : k = m, \dots, n\} = \{m, \dots, n\}.$$

Thus π is uniquely determined if $m = 1$.

Now assume that $m > 1$. As $\pi \in S_n$, we must have

$$\{\pi(k) : k = 1, \dots, m-1\} = \{1, \dots, m-1\}.$$

Since $k + \pi(k)$ is a power of two for every $k = 1, \dots, m-1$, by the induction hypothesis we have $\pi(k) = \pi_m(k)$ for all $k = 1, \dots, m-1$. Thus π is indeed uniquely determined.

In view of the above, the proof of Theorem 1.2 is now complete. $\square$

Proof of Corollary 1.3. Clearly, $\pi \in S_{2n}$ and $k + \pi(k) \in \{2^a - 1 : a \in \mathbb{Z}^+\}$ for all $k = 1, \dots, 2n$, if and only if there are $\sigma, \tau \in S_n$ with $\pi(2k) = 2\sigma(k) - 1$ and $\pi(2k-1) = 2\tau(k)$ for all $k = 1, \dots, n$ such that $k + \sigma(k), k + \tau(k) \in \{2^a - 1 : a \in \mathbb{Z}^+\}$ for all $k = 1, \dots, n$. Thus we get the desired result by applying Theorem 1.2. $\square$

Lemma 2.1 (Alon's Combinatorial Nullstellensatz [A]). *Let $A_1, \dots, A_n$ be finite subsets of a field F with $|A_i| > k_i$ for $i = 1, \dots, n$ where $k_1, \dots, k_n \in \{0, 1, 2, \dots\}$. If the coefficient of the monomial $x_1^{k_1} \dots x_n^{k_n}$ in $P(x_1, \dots, x_n) \in F[x_1, \dots, x_n]$ is nonzero and $k_1 + \dots + k_n$ is the total degree of P , then there are $a_1 \in A_1, \dots, a_n \in A_n$ such that $P(a_1, \dots, a_n) \neq 0$.*

Lemma 2.2. *Let $a_1, \dots, a_n$ be elements of a field F . Then the coefficient of $x_1^{n-1} \dots x_n^{n-1}$ in the polynomial*

$$\prod_{1 \leq i < j \leq n} (x_j - x_i)(a_j x_j - a_i x_i) \in F[x_1, \dots, x_n]$$

is $(-1)^{n(n-1)/2} \text{per}[a_i^{j-1}]_{1 \leq i, j \leq n}$.

Proof. This is easy. In fact,

$$\begin{aligned} & \prod_{1 \leq i < j \leq n} (x_j - x_i)(a_j x_j - a_i x_i) \\ &= (-1)^{\binom{n}{2}} \det[x_i^{n-j}]_{1 \leq i, j \leq n} \times \det[a_i^{j-1} x_i^{j-1}]_{1 \leq i, j \leq n} \\ &= (-1)^{\binom{n}{2}} \sum_{\sigma \in S_n} \text{sign}(\sigma) \prod_{i=1}^n x_i^{n-\sigma(i)} \sum_{\tau \in S_n} \text{sign}(\tau) \prod_{i=1}^n a_i^{\tau(i)-1} x_i^{\tau(i)-1}. \end{aligned}$$

Therefore the coefficient of $x_1^{n-1} \dots x_n^{n-1}$ in this polynomial is

$$(-1)^{\binom{n}{2}} \sum_{\sigma \in S_n} \text{sign}(\sigma)^2 \prod_{i=1}^n a_i^{\sigma(i)-1} = (-1)^{n(n-1)/2} \text{per}[a_i^{j-1}]_{1 \leq i, j \leq n}.$$

This concludes the proof. $\square$

Remark 2.1. See [DKSS] and [S08, Lemma 2.2] for similar identities and arguments.

Proof of Theorem 1.3. (i) Let g be a primitive root modulo p . Then, there is a permutation $\pi \in S_{p-1}$ such that the numbers $k\pi(k)$ ($k = 1, \dots, p-1$) are pairwise incongruent modulo p , if and only if there is a permutation $\rho \in S_{p-1}$ such that $g^{i+\rho(i)}$ ($i = 1, \dots, p-1$) are pairwise incongruent modulo p (i.e., the numbers $i + \rho(i)$ ($i = 1, \dots, p-1$) are pairwise incongruent modulo $p-1$).

Suppose that $\rho \in S_{p-1}$ and all the numbers $i + \rho(i)$ ($i = 1, \dots, p-1$) are pairwise incongruent modulo $p-1$. Then

$$\sum_{i=1}^{p-1} (i + \rho(i)) \equiv \sum_{j=1}^{p-1} j \pmod{p-1},$$

and hence $\sum_{i=1}^{p-1} i = p(p-1)/2 \equiv 0 \pmod{p-1}$ which is impossible. This contradiction proves the first assertion in Theorem 1.3(i).

Now we turn to prove the second assertion in Theorem 1.3(i). Suppose that $\text{per}[i^{j-1}]_{1 \leq i, j \leq p-1} \not\equiv 0 \pmod{p}$. Then, by Lemma 2.2, the coefficient of $x_1^{p-2} \dots x_{p-1}^{p-2}$ in the polynomial

$$\prod_{1 \leq i < j \leq p-1} (x_j - x_i)(jx_j - ix_i)$$

is not congruent to zero modulo p . Applying Lemma 2.1 with $F = \mathbb{Z}/p\mathbb{Z}$ and $A = \{k + p\mathbb{Z} : k = 1, \dots, p-1\}$, we see that there is a permutation $\pi \in S_{p-1}$ such that all those $k\pi(k)$ ($k = 1, \dots, p-1$) are pairwise incongruent modulo p , which contradicts the first assertion of Theorem 1.3(i) we have just proved.

(ii) Let $n > 2$ be an integer. Then

$$\begin{aligned} \text{per}[i^{j-1}]_{1 \leq i, j \leq n} &= \sum_{\sigma \in S_n} \prod_{k=1}^n k^{\sigma(k)-1} \\ &\equiv \sum_{\substack{\sigma \in S_n \\ \sigma(n)=1}} (n-1)! \prod_{k=1}^{n-1} k^{\sigma(k)-2} = (n-1)! \sum_{\tau \in S_{n-1}} \prod_{k=1}^{n-1} k^{\tau(k)-1} \\ &= (n-1)! \text{per}[i^{j-1}]_{1 \leq i, j \leq n-1} \pmod{n}. \end{aligned}$$

We want to prove that $n \mid \text{per}[i^{j-1}]_{1 \leq i, j \leq n}$. This holds when n is an odd prime p , because $p \mid \text{per}[i^{j-1}]_{1 \leq i, j \leq p-1}$ by Theorem 1.3(i). For $n = 4$, we have

$$\begin{aligned} \text{per}[i^{j-1}]_{1 \leq i, j \leq 4} &\equiv 3! \sum_{\tau \in S_3} 1^{\tau(1)-1} 2^{\tau(2)-1} 3^{\tau(3)-1} \\ &\equiv 6 (1^{2-1} 2^{1-1} 3^{3-1} + 1^{3-1} 2^{1-1} 3^{2-1}) \equiv 0 \pmod{4}. \end{aligned}$$

Now assume that $n > 4$ is composite. By the above, it suffices to show that $(n-1)! \equiv 0 \pmod{n}$. Let p be the smallest prime divisor of n . Then $n = pq$ for some integer $q \geq p$. If $p < q$, then $n = pq$ divides $(n-1)!$. If $q = p$, then $p^2 = n > 4$ and hence $2p < p^2$, thus $2n = p(2p)$ divides $(n-1)!$.

In view of the above, we have completed the proof of Theorem 1.3. $\square$

3. PROOFS OF THEOREMS 1.4 AND 1.5

Proof of Theorem 1.4. (i) The case $n = 1$ is trivial. Below we let $n > 1$. Note that the subgroup H of G generated by $a_1, \dots, a_n$ is infinite, finitely generated and torsion-free. Thus H is isomorphic to $\mathbb{Z}^r$ for some positive integer r . By algebraic number theory (cf. [He]), we may take an algebraic number field K with $[K : \mathbb{Q}] = r$ and hence H is isomorphic to the additive group O_K of

algebraic integers in K . Thus, without any loss of generality, we may simply assume that G is the additive group $\mathbb{C}$ of all complex numbers.

By Lemma 2.2, the coefficient of $x_1^{n-1} \dots x_n^{n-1}$ in the polynomial

$$P(x_1, \dots, x_n) := \prod_{1 \leq i < j \leq n} (x_j - x_i)(jx_j - ix_i) \in \mathbb{C}[x_1, \dots, x_n]$$

is $(-1)^{n(n-1)/2} \text{per}[i^{j-1}]_{1 \leq i, j \leq n}$, which is nonzero since $\text{per}[i^{j-1}]_{1 \leq i, j \leq n} > 0$. Applying Lemma 2.1 we see that there are $x_1, \dots, x_n \in A = \{a_1, \dots, a_n\}$ with $P(x_1, \dots, x_n) \neq 0$. Thus, for some $\pi \in S_n$ all the numbers $ka_{\sigma(k)}$ ($k = 1, \dots, n$) are distinct. This ends the proof of part (i).

(ii) Let e be the identity of the group G . Suppose that $a = b^2$ and also $a^2 = b$. Then $a = (a^2)^2 = a^4$, and hence $a^3 = e$. As the order of a is not three, we have $a = e$ and hence $b = a^2 = e$, which leads to a contradiction since $a \neq b$. Therefore $a^{\sigma(1)}$ and $b^{\sigma(2)}$ are distinct for some $\sigma \in S_2$.

To prove the second assertion in Theorem 1.4(ii), we distinguish two cases.

Case 1. One of a, b, c is the square of another element among a, b, c .

Without loss of generality, we simply assume that $a = b^2$. As $a \neq b$ we have $b \neq e$. As b is not of order two, we also have $a \neq e$. Note that $b^2 = a \neq c$. If $b^2 = a^3$, then $a = a^3$ which is impossible since the order of a is not two. If $a^3 \neq c$, then c, b^2, a^3 are pairwise distinct.

Now assume that $a^3 = c$. As a is not of order three, we have $b \neq a^2$ and $c \neq e$. Note that $a^3 = c \neq b$ and also $a^3 = c \neq c^2$. If $b \neq c^2$, then b, c^2, a^3 are pairwise distinct. If $b = c^2$, then $a = b^2 = c^4 = (a^3)^4$ and hence the order of a is 11, thus $a^2 \neq (a^3)^3 = c^3$ and hence b, a^2, c^3 are pairwise distinct.

Case 2. None of a, b, c is the square of another one among a, b, c .

Suppose that there is no $\tau \in S_3$ with $a^{\tau(1)}, b^{\tau(2)}, c^{\tau(3)}$ pairwise distinct. Then $c^3 \in \{a, b^2\} \cap \{a^2, b\}$. If $c^3 = a$, then $c^3 \neq b$ and hence $a = c^3 = a^2$, thus $a = e = c$ which leads to a contradiction. (Recall that none of a, b, c is of order 3.) Therefore $c^3 = b^2$. As c is not of order three, if $b = e$ then we have $c = e = b$ which is impossible. So $c^3 = b^2 \neq b$ and hence $b^2 = c^3 = a^2$. Similarly, $a^3 = b^2 = c^2$. Thus $a^3 = b^2 = a^2$, hence $a = e$ and $b^2 = a^2 = e$, which contradicts $b \neq a$ since b is not of order two.

In view of the above, we have finished the proof of Theorem 1.4. $\square$

Proof of Theorem 1.5. The subgroup of G generated by $a_1, \dots, a_n$ is a finitely generated torsion-free abelian group. So we may simply assume that $G = \mathbb{Z}^r$ for some positive integer r without any loss of generality. It is well known that there is a linear ordering $\leq$ on $G = \mathbb{Z}^r$ such that for any $a, b, c \in G$ if $a < b$ then $-b < -a$ and $a + c < b + c$ (cf. [L]). For convenience, we suppose $a_1 < a_2 < \dots < a_n$ without any loss of generality.

If $n = 4$, then $(b_1, b_2, b_3, b_4) = (a_1, a_2, a_3, a_4)$ meets the requirement since

$$a_1 + a_2 + a_3 < a_4 + a_1 + a_2 < a_3 + a_4 + a_1 < a_2 + a_3 + a_4.$$

Below we assume $n \geq 5$.

Clearly

$$a_1 + a_2 + a_3 < a_2 + a_3 + a_4 < \dots < a_{n-2} + a_{n-1} + a_n.$$

For convenience we set

$$S := \{a_{i-1} + a_i + a_{i+1} : i = 2, \dots, n-1\},$$

and let $\min S$ and $\max S$ denote the least element and the largest element of S respectively. Note that

$$\min S = a_1 + a_2 + a_3 < a_n + a_1 + a_2 < a_{n-1} + a_n + a_1 < \max S = a_{n-2} + a_{n-1} + a_n.$$

If $\{a_n + a_1 + a_2, a_{n-1} + a_n + a_1\} \cap S = \emptyset$, then $(b_1, \dots, b_n) = (a_1, \dots, a_n)$ meets the requirement. Obviously

$$-a_n < -a_{n-1} < \dots < -a_2 < -a_1 \text{ and } (-a_2) + (-a_1) + (-a_n) = -(a_1 + a_2 + a_n).$$

So, it suffices to find a desired permutation $b_1, \dots, b_n$ of $a_1, \dots, a_n$ under the condition $a_{n-1} + a_n + a_1 \in S$.

Case 1. $n = 5$.

As $a_4 + a_5 + a_1 \in S$, we have $a_4 + a_5 + a_1 = a_2 + a_3 + a_4$ and we may take $(b_1, \dots, b_5) = (a_1, a_2, a_3, a_5, a_4)$ since

$$a_1 + a_2 + a_3 < a_4 + a_1 + a_2 < a_2 + a_3 + a_4 = a_5 + a_4 + a_1 < a_2 + a_3 + a_5 < a_3 + a_5 + a_4.$$

Case 2. $n = 6$.

As $a_5 + a_6 + a_1 \in S$, the sum $a_5 + a_6 + a_1$ is equal to $a_2 + a_3 + a_4$ or $a_3 + a_4 + a_5$. If $a_5 + a_6 + a_1 = a_2 + a_3 + a_4$, then we may take $(b_1, \dots, b_6) = (a_1, a_2, a_5, a_3, a_4, a_6)$ since

$$\begin{aligned} a_1 + a_2 + a_5 &< a_6 + a_1 + a_2 < a_4 + a_6 + a_1 < a_5 + a_6 + a_1 = a_2 + a_3 + a_4 \\ &< a_2 + a_5 + a_3 < a_5 + a_3 + a_4 < a_3 + a_4 + a_6. \end{aligned}$$

If $a_5 + a_6 + a_1 = a_3 + a_4 + a_5$, then $a_6 + a_1 = a_3 + a_4$ and we may take $(b_1, \dots, b_6) = (a_1, a_2, a_3, a_4, a_6, a_5)$ since

$$\begin{aligned} a_1 + a_2 + a_3 &< a_5 + a_1 + a_2 < a_6 + a_1 + a_2 = a_2 + a_3 + a_4 \\ &< a_3 + a_4 + a_5 = a_6 + a_5 + a_1 < a_3 + a_4 + a_6 < a_4 + a_6 + a_5. \end{aligned}$$

Case 3. $n = 7$.

As $a_6 + a_7 + a_1 \in S$, the sum $a_6 + a_7 + a_1$ is equal to $a_2 + a_3 + a_4$ or $a_3 + a_4 + a_5$ or $a_4 + a_5 + a_6$. If $a_6 + a_7 + a_1 = a_4 + a_5 + a_6$, then $a_7 + a_1 = a_4 + a_5$ and we may take $(b_1, \dots, b_7) = (a_2, a_1, a_4, a_5, a_3, a_6, a_7)$ since

$$\begin{aligned} a_2 + a_1 + a_4 &< a_1 + a_4 + a_5 = a_1 + a_1 + a_7 < a_7 + a_2 + a_1 \\ &< a_7 + a_1 + a_3 = a_4 + a_5 + a_3 < a_5 + a_3 + a_6 \\ &< a_4 + a_5 + a_6 = a_1 + a_6 + a_7 < a_2 + a_6 + a_7 < a_3 + a_6 + a_7. \end{aligned}$$

If $a_6 + a_7 + a_1 = a_2 + a_3 + a_4$, then we may take $(b_1, \dots, b_7) = (a_1, a_2, a_3, a_5, a_4, a_6, a_7)$ since

$$\begin{aligned} a_1 + a_2 + a_3 &< a_7 + a_1 + a_2 < a_5 + a_7 + a_1 < a_6 + a_7 + a_1 = a_2 + a_3 + a_4 \\ &< a_2 + a_3 + a_5 < a_3 + a_5 + a_4 < a_5 + a_4 + a_6 < a_4 + a_6 + a_7. \end{aligned}$$

If $a_6 + a_7 + a_1 = a_3 + a_4 + a_5$ and $a_5 + a_6 + a_1 \neq a_2 + a_3 + a_4$, then $a_6 + a_1 < a_3 + a_4$ and we may take $(b_1, \dots, b_7) = (a_1, a_2, a_3, a_4, a_7, a_5, a_6)$ since

$$\begin{aligned} a_1 + a_2 + a_3 &< a_6 + a_1 + a_2 < \min\{a_5 + a_6 + a_1, a_2 + a_3 + a_4\} \\ &< \max\{a_5 + a_6 + a_1, a_2 + a_3 + a_4\} < a_1 + a_6 + a_7 = a_3 + a_4 + a_5 \\ &< a_3 + a_4 + a_7 < a_4 + a_7 + a_5 < a_7 + a_5 + a_6. \end{aligned}$$

If $a_6 + a_7 + a_1 = a_3 + a_4 + a_5$ and $a_5 + a_6 + a_1 = a_2 + a_3 + a_4$, then $a_7 + a_1 < a_3 + a_4$ and we may take $(b_1, \dots, b_7) = (a_1, a_2, a_3, a_4, a_6, a_5, a_7)$ since

$$\begin{aligned} a_1 + a_2 + a_3 &< a_7 + a_1 + a_2 < a_5 + a_6 + a_1 = a_2 + a_3 + a_4 \\ &< a_5 + a_7 + a_1 < a_3 + a_4 + a_5 = a_6 + a_7 + a_1 \\ &< a_3 + a_4 + a_6 < a_4 + a_6 + a_5 < a_6 + a_5 + a_7. \end{aligned}$$

Case 4. $n > 7$ and $a_n + a_1 + a_2 \notin S$.

In this case, there is a unique $2 < i < n - 1$ with $a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1$. If $i < n - 3$, then we may take

$$(b_1, \dots, b_n) = (a_1, \dots, a_{i-2}, a_{i-1}, a_i, a_{i+2}, a_{i+1}, a_{i+3}, \dots, a_n)$$

because

$$\begin{aligned} a_{i-2} + a_{i-1} + a_i &< a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1 < a_{i-1} + a_i + a_{i+2} \\ &< a_i + a_{i+2} + a_{i+1} < a_{i+2} + a_{i+1} + a_{i+3} \\ &< a_{i+1} + a_{i+3} + a_{i+4} < \dots < a_{n-2} + a_{n-1} + a_n. \end{aligned}$$

When $i \in \{n - 2, n - 3\}$, we have $i \geq n - 3 > 4$, and hence in the case $a_1 + a_2 + a_n \neq a_{i-4} + a_{i-3} + a_{i-1}$ we may take

$$(b_1, \dots, b_n) = (a_1, \dots, a_{i-4}, a_{i-3}, a_{i-1}, a_{i-2}, a_i, a_{i+1}, a_{i+2}, \dots, a_n)$$

because

$$\begin{aligned}
a_{i-4} + a_{i-3} + a_{i-2} &< a_{i-4} + a_{i-3} + a_{i-1} < a_{i-3} + a_{i-1} + a_{i-2} \\
&< a_{i-1} + a_{i-2} + a_i < a_{i-2} + a_i + a_{i+1} \\
&< a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1 \\
&< a_i + a_{i+1} + a_{i+2} < \dots < a_{n-2} + a_{n-1} + a_n
\end{aligned}$$

and

$$\begin{aligned}
a_n + a_1 + a_2 &< (a_{i-2} + a_{n-1} - a_{i+1}) + a_n + a_1 \\
&= a_{i-2} - a_{i+1} + (a_{i-1} + a_i + a_{i+1}) = a_{i-1} + a_{i-2} + a_i.
\end{aligned}$$

If $i \in \{n-2, n-3\}$ and $a_1 + a_2 + a_n = a_{i-4} + a_{i-3} + a_{i-1}$, then we may take

$$(b_1, \dots, b_n) = (a_1, \dots, a_{i-4}, a_{i-3}, a_i, a_{i-2}, a_{i-1}, a_{i+1}, a_{i+2}, \dots, a_n)$$

because

$$\begin{aligned}
a_n + a_1 + a_2 &= a_{i-4} + a_{i-3} + a_{i-1} \\
&< a_{i-4} + a_{i-3} + a_i < a_{i-3} + a_i + a_{i-2} < a_i + a_{i-2} + a_{i-1} \\
&< a_{i-2} + a_{i-1} + a_{i+1} < a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1 \\
&< a_{i-1} + a_{i+1} + a_{i+2} < \dots < a_{n-2} + a_{n-1} + a_n.
\end{aligned}$$

Case 5. $n > 7$ and $a_n + a_1 + a_2 \in S$.

In this case, for some $2 < j < i \leq n-2$ we have

$$a_{n-1} + a_n + a_1 = a_{i-1} + a_i + a_{i+1} > a_{j-1} + a_j + a_{j+1} = a_n + a_1 + a_2.$$

If $j+1 = i$, then

$$\begin{aligned}
a_{n-1} - a_2 &= (a_{n-1} + a_n + a_1) - (a_n + a_1 + a_2) \\
&= a_{i-1} + a_i + a_{i+1} - (a_i + a_{i-1} + a_{i-2}) = a_{i+1} - a_{i-2}
\end{aligned}$$

which is impossible since $i \geq 4$ and $n > 6$.

If $i-j > 5$, then

$$(b_1, \dots, b_n) = (a_1, \dots, a_{j-1}, a_j, a_{j+2}, a_{j+1}, a_{j+3}, \dots, a_{i-3}, a_{i-1}, a_{i-2}, a_i, a_{i+1}, \dots, a_n)$$

meets the requirement since

$$\begin{aligned}
a_{j-1} + a_j + a_{j+1} &= a_n + a_1 + a_2 < a_{j-1} + a_j + a_{j+2} \\
&< a_j + a_{j+2} + a_{j+1} < a_{j+2} + a_{j+1} + a_{j+3} \\
&< \dots < a_{i-3} + a_{i-1} + a_{i-2} < a_{i-1} + a_{i-2} + a_i \\
&< a_{i-2} + a_i + a_{i+1} < a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1 \\
&< a_i + a_{i+1} + a_{i+2} < \dots < a_{n-2} + a_{n-1} + a_n.
\end{aligned}$$

If $i - j = 5$, then $j + 4 = i - 1$ and

$$(b_1, \dots, b_n) = (a_1, \dots, a_{j-1}, a_j, a_{j+2}, a_{j+1}, a_{i-1}, a_{i-2}, a_i, a_{i+1}, \dots, a_n)$$

meets the requirement. If $i - j = 4$, then

$$(b_1, \dots, b_n) = (a_1, \dots, a_{j-1}, a_j, a_{j+2}, a_{j+3}, a_{j+1}, a_i, a_{i+1}, \dots, a_n)$$

meets the requirement since

$$\begin{aligned} a_{j-1} + a_j + a_{j+1} &= a_n + a_1 + a_2 \\ &< a_{j-1} + a_j + a_{j+2} < a_j + a_{j+2} + a_{j+3} \\ &< a_{j+2} + a_{j+3} + a_{j+1} < a_{j+3} + a_{j+1} + a_i \\ &< a_{j+1} + a_i + a_{i+1} < a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1 \\ &< a_i + a_{i+1} + a_{i+2} < \dots < a_{n-2} + a_{n-1} + a_n. \end{aligned}$$

If $i - j = 3$, then

$$(b_1, \dots, b_n) = (a_1, \dots, a_{j-1}, a_j, a_{j+2}, a_{j+1}, a_i, a_{i+1}, \dots, a_n)$$

meets the requirement since

$$\begin{aligned} a_{j-1} + a_j + a_{j+1} &= a_n + a_1 + a_2 \\ &< a_{j-1} + a_j + a_{j+2} < a_j + a_{j+2} + a_{j+1} \\ &< a_{j+2} + a_{j+1} + a_i = a_{i-1} + a_{i-2} + a_i < a_{i-2} + a_i + a_{i+1} \\ &< a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1 \\ &< a_i + a_{i+1} + a_{i+2} < \dots < a_{n-2} + a_{n-1} + a_n. \end{aligned}$$

If $j > 4$ and $i = j + 2$, then

$$(b_1, \dots, b_n) = (a_1, \dots, a_{j-3}, a_{j-1}, a_{j-2}, a_{j+1}, a_j, a_i, a_{i+1}, a_{i+2}, \dots, a_n)$$

meets the requirement since

$$\begin{aligned} a_{j-4} + a_{j-3} + a_{j-1} &< a_{j-3} + a_{j-1} + a_{j-2} < a_{j-1} + a_{j-2} + a_{j+1} \\ &< a_{j-2} + a_{j+1} + a_j < a_{j-1} + a_j + a_{j+1} = a_n + a_1 + a_2 \\ &< a_{j+1} + a_j + a_i < a_j + a_i + a_{i+1} \\ &< a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1 < a_i + a_{i+1} + a_{i+2}. \end{aligned}$$

If $i = j + 2 \leq n - 4$, then

$$(b_1, \dots, b_n) = (a_1, \dots, a_{j-2}, a_{j-1}, a_j, a_i, a_{i-1}, a_{i+2}, a_{i+1}, a_{i+3}, a_{i+4}, \dots, a_n)$$

meets the requirement since

$$\begin{aligned}
a_{j-2} + a_{j-1} + a_j &< a_{j-1} + a_j + a_{j+1} = a_n + a_1 + a_2 \\
&< a_{j-1} + a_j + a_i < a_j + a_i + a_{i-1} \\
&< a_{i-1} + a_i + a_{i+1} = a_{n-1} + a_n + a_1 \\
&< a_i + a_{i-1} + a_{i+2} < a_{i-1} + a_{i+2} + a_{i+1} \\
&< a_{i+2} + a_{i+1} + a_{i+3} < a_{i+1} + a_{i+3} + a_{i+4} \\
&< \dots < a_{n-2} + a_{n-1} + a_n.
\end{aligned}$$

If $i \geq n-3$, $j \leq 4$ and $i-j = 2$, then $2 = i-j \geq n-3-4$ and hence $n \in \{8, 9\}$.

For $n = 8$, we need to consider the case $i = 6$ and $j = 4$. As $a_8 + a_1 + a_2 = a_3 + a_4 + a_5$ and $a_7 + a_8 + a_1 = a_5 + a_6 + a_7$, we have $a_8 + a_1 = a_5 + a_6 = a_3 + a_4 + a_5 - a_2$. If $2a_5 \neq a_4 + a_7$, then $a_5 + a_8 + a_1 = 2a_5 + a_6 \neq a_4 + a_6 + a_7$ and hence we may take

$$(b_1, \dots, b_8) = (a_1, a_2, a_3, a_4, a_6, a_7, a_5, a_8)$$

since

$$\begin{aligned}
a_1 + a_2 + a_3 &< a_2 + a_3 + a_4 < a_3 + a_4 + a_5 = a_8 + a_1 + a_2 < a_3 + a_4 + a_6 \\
&< \min\{a_4 + a_6 + a_7, a_5 + a_8 + a_1\} < \max\{a_4 + a_6 + a_7, a_5 + a_8 + a_1\} \\
&< a_6 + a_7 + a_5 = a_7 + a_8 + a_1 < a_7 + a_5 + a_8.
\end{aligned}$$

If $2a_5 = a_4 + a_7$, then $a_6 + a_8 + a_1 = a_5 + 2a_6 > a_4 + a_5 + a_7$ and we may take

$$(b_1, \dots, b_8) = (a_1, a_2, a_3, a_4, a_5, a_7, a_8, a_6)$$

since

$$\begin{aligned}
a_1 + a_2 + a_3 &< a_1 + a_3 + a_4 = a_1 + a_2 + a_6 < a_2 + a_3 + a_4 \\
&< a_3 + a_4 + a_5 = a_8 + a_1 + a_2 < a_4 + a_5 + a_7 < a_6 + a_8 + a_1 \\
&< a_5 + a_7 + a_8 < a_7 + a_8 + a_6.
\end{aligned}$$

When $n = 8$, $i = 5$ and $j = 3$, it suffices to apply the result for $i = 6$ and $j = 4$ to the sequence

$$\begin{aligned}
a'_1 = -a_8 &< a'_2 = -a_7 < a'_3 = -a_6 < a'_4 = -a_5 \\
&< a'_5 = -a_4 < a'_6 = -a_3 < a'_7 = -a_2 < a'_8 = -a_1
\end{aligned}$$

since $a'_7 + a'_8 + a'_1 = -(a_1 + a_2 + a_8) = -(a_2 + a_3 + a_4) = a'_5 + a'_6 + a'_7$ and $a'_8 + a'_1 + a'_2 = -(a_1 + a_7 + a_8) = -(a_4 + a_5 + a_6) = a'_3 + a'_4 + a'_5$.

Now it remains to consider the last case where $n = 9$, $i = 6$ and $j = 4$. As $a_3 + a_4 + a_5 = a_9 + a_1 + a_2$ and $a_5 + a_6 + a_7 = a_8 + a_9 + a_1$, we have $a_3 + a_4 < a_9 + a_1$ and hence $a_3 + a_4 + a_6 < a_3 + a_4 + a_7 < a_7 + a_9 + a_1$. If $a_7 + a_9 + a_1 = a_4 + a_5 + a_6$, then

$$a_8 - a_7 = (a_8 + a_9 + a_1) - (a_7 + a_9 + a_1) = a_5 + a_6 + a_7 - (a_4 + a_5 + a_6) = a_7 - a_4.$$

When $2a_7 \neq a_8 + a_4$, we have $a_7 + a_9 + a_1 \neq a_4 + a_5 + a_6$ and hence we may take

$$(b_1, \dots, b_9) = (a_1, a_2, a_3, a_4, a_6, a_5, a_8, a_7, a_9)$$

since

$$\begin{aligned} a_1 + a_2 + a_3 &< a_2 + a_3 + a_4 < a_3 + a_4 + a_5 = a_9 + a_1 + a_2 < a_3 + a_4 + a_6 \\ &< \min\{a_4 + a_5 + a_6, a_7 + a_9 + a_1\} < \max\{a_4 + a_5 + a_6, a_7 + a_9 + a_1\} \\ &< a_6 + a_5 + a_7 = a_8 + a_9 + a_1 < a_6 + a_5 + a_8 \\ &< a_5 + a_8 + a_7 < a_8 + a_7 + a_9. \end{aligned}$$

If $2a_7 = a_8 + a_4$, then $a_5 + a_6 + a_7 < 2a_7 + a_6 = a_4 + a_6 + a_8$ and hence we may take

$$(b_1, \dots, b_9) = (a_1, a_2, a_3, a_4, a_6, a_8, a_5, a_7, a_9)$$

since

$$\begin{aligned} a_1 + a_2 + a_3 &< a_2 + a_3 + a_4 < a_3 + a_4 + a_5 = a_9 + a_1 + a_2 < a_3 + a_4 + a_6 \\ &< a_9 + a_1 + a_6 < a_7 + a_9 + a_1 < a_8 + a_9 + a_1 = a_5 + a_6 + a_7 \\ &< a_4 + a_6 + a_8 < a_6 + a_8 + a_5 < a_8 + a_5 + a_7 < a_5 + a_7 + a_9. \end{aligned}$$

In view of the above, we have completed the proof of Theorem 1.5. $\square$

4. SOME CONJECTURES

Motivated by Theorem 1.3(i) and Theorem 1.4, we pose the following conjecture for finite groups.

Conjecture 4.1. *Let n be a positive integer, and let G be a group containing no element of order among $2, \dots, n+1$. Then, for any $A \subseteq G$ with $|A| = n$, we may write $A = \{a_1, \dots, a_n\}$ with $a_1, a_2^2, \dots, a_n^n$ pairwise distinct.*

Remark 4.1. (a) Theorem 1.4 shows that this conjecture holds when $n \leq 3$ or G is a torsion-free abelian group.

(b) For $n = 4, 5, 6, 7, 8, 9$ we have verified the conjecture for cyclic groups $G = \mathbb{Z}/m\mathbb{Z}$ with $|G| = m$ not exceeding 100, 100, 70, 60, 30, 30 respectively.

(c) If G is a finite group with $|G| > 1$, then the least order of a non-identity element of G is $p(G)$, the smallest prime divisor of $|G|$.

Inspired by Theorem 1.3, we formulate the following conjecture.

Conjecture 4.2. *Let $n > 1$ be an integer with $n \not\equiv 2 \pmod{4}$.*

(i) *We have*

$$\text{per}[i^{j-1}]_{1 \leq i, j \leq n-1} \equiv 0 \pmod{n}. \quad (4.1)$$

(ii) *If $n \equiv 1 \pmod{3}$, then*

$$\text{per}[i^{j-1}]_{1 \leq i, j \leq n-1} \equiv 0 \pmod{n^2}. \quad (4.2)$$

Remark 4.2. We have checked this conjecture via computing $\text{per}[i^{j-1}]_{1 \leq i, j \leq n-1}$ modulo n^2 for $n \leq 17$. The sequence $a_n = \text{per}[i^{j-1}]_{1 \leq i, j \leq n}$ ($n = 1, 2, 3, \dots$) is available from [S18, A322363].

Conjecture 4.3. (i) *For any $n \in \mathbb{Z}^+$, there is a permutation $\sigma_n \in S_n$ such that $k\sigma_n(k) + 1$ is prime for every $k = 1, \dots, n$.*

(ii) *For any integer $n > 2$, there is a permutation $\tau_n \in S_n$ such that $k\tau_n(k) - 1$ is prime for every $k = 1, \dots, n$.*

Remark 4.3. See [S18, A321597] for related data and examples.

Conjecture 4.4. (i) *For each $n \in \mathbb{Z}^+$, there is a permutation π_n of $\{1, \dots, n\}$ such that $k^2 + k\pi_n(k) + \pi_n(k)^2$ is prime for every $k = 1, \dots, n$.*

(ii) *For any positive integer $n \neq 7$, there is a permutation π_n of $\{1, \dots, n\}$ such that $k^2 + \pi_n(k)^2$ is prime for every $k = 1, \dots, n$.*

Remark 4.4. See [S18, A321610] for related data and examples.

As usual, for $k = 1, 2, 3, \dots$ we let p_k denote the k -th prime.

Conjecture 4.5. *For any $n \in \mathbb{Z}^+$, there is a permutation $\pi \in S_n$ such that $p_k + p_{\pi(k)} + 1$ is prime for every $k = 1, \dots, n$.*

Remark 4.5. See [S18, A321727] for related data and examples.

In 1973 J.-R. Chen [Ch] proved that there are infinitely many primes p with $p+2$ a product of at most two primes; nowadays such primes p are called Chen primes.

Conjecture 4.6. *Let $n \in \mathbb{Z}^+$. Then, there is an even permutation $\sigma \in S_n$ with $p_k p_{\sigma(k)} - 2$ prime for all $k = 1, \dots, n$. If $n > 2$, then there is an odd permutation $\tau \in S_n$ with $p_k p_{\tau(k)} - 2$ prime for all $k = 1, \dots, n$.*

Remark 4.6. See [S18, A321855] for related data and examples. If we let $b(n)$ denote the number of even permutations $\sigma \in S_n$ with $p_k p_{\sigma(k)} - 2$ prime for all $k = 1, \dots, n$, then

$$(b(1), \dots, b(11)) = (1, 1, 1, 1, 3, 6, 1, 1, 33, 125, 226).$$

Conjecture 2.17(ii) of Sun [S15] implies that for any odd integer $n > 1$ there is a prime $p \leq n$ such that $pn - 2$ is prime.

In 2002, Cloitre [Cl, A073112] created the sequence A073112 on OEIS whose n -th term is the number of permutations $\pi \in S_n$ with $\sum_{k=1}^n \frac{1}{k+\pi(k)} \in \mathbb{Z}$. Recently Sun [S18a] conjectured that for any integer $n > 5$ there is a permutation $\pi \in S_n$ satisfying

$$\sum_{k=1}^n \frac{1}{k + \pi(k)} = 1,$$

and this was later confirmed by the user Zhao Shen at Mathoverflow via clever induction arguments.

In 1982 A. Filz (cf. [G, pp. 160-162]) conjectured that for any $n = 2, 4, 6, \dots$ there is a circular permutation $(i_1, \dots, i_n)$ of $1, \dots, n$ such that all the n adjacent sums

$$i_1 + i_2, i_2 + i_3, \dots, i_{n-1} + i_n, i_n + i_1$$

are prime.

Motivated by this, we pose the following conjecture.

Conjecture 4.7. (i) *For any integer $n > 6$, there is a permutation $\pi \in S_n$ such that*

$$\sum_{k=1}^{n-1} \frac{1}{\pi(k) + \pi(k+1)} = 1. \quad (4.3)$$

Also, for any integer $n > 7$, there is a permutation $\pi \in S_n$ such that

$$\frac{1}{\pi(1) + \pi(2)} + \frac{1}{\pi(2) + \pi(3)} + \dots + \frac{1}{\pi(n-1) + \pi(n)} + \frac{1}{\pi(n) + \pi(1)} = 1. \quad (4.4)$$

(ii) *For any integer $n > 7$, there is a permutation $\pi \in S_n$ such that*

$$\sum_{k=1}^{n-1} \frac{1}{\pi(k)^2 - \pi(k+1)^2} = 0. \quad (4.5)$$

Remark 4.7. See [S18, A322070 and A322099] for related data and examples. For the latter assertion in Conjecture 4.7(i), the equality (4.4) with $n = 8$ holds if we take $(\pi(1), \dots, \pi(8)) = (6, 1, 5, 2, 4, 3, 7, 8)$. In a previous version of this paper posted to arXiv, the author also conjectured that for any integer $n > 5$ there is a permutation $\pi \in S_n$ with $\sum_{k=1}^{n-1} \frac{1}{\pi(k)\pi(k+1)} = 1$; this, together with two other conjectures of the author, was confirmed by G.-N. Han [H].

Conjecture 4.8. (i) *For any integer $n > 1$, there is a permutation $\pi \in S_n$ such that*

$$\sum_{0 < k < n} \pi(k)\pi(k+1) \in \{2^m + 1 : m = 0, 1, 2, \dots\}. \quad (4.6)$$

(ii) *For any integer $n > 4$, there is a unique power of two which can be written as $\sum_{k=1}^{n-1} \pi(k)\pi(k+1)$ with $\pi \in S_n$ and $\pi(n) = n$.*

Remark 4.8. Concerning part (i) of Conjecture 4.8, when $n = 4$ we may choose $(\pi(1), \dots, \pi(4)) = (1, 3, 2, 4)$ so that

$$\sum_{k=1}^3 \pi(k)\pi(k+1) = 1 \times 3 + 3 \times 2 + 2 \times 4 = 2^4 + 1.$$

For any $\pi \in S_n$, if for each $k = 1, \dots, n$ we let

$$\pi'(k) = \begin{cases} \pi(\pi^{-1}(k) + 1) & \text{if } \pi^{-1}(k) \neq n, \\ \pi(1) & \text{if } \pi^{-1}(k) = n, \end{cases}$$

then $\pi' \in S_n$ and

$$\pi(1)\pi(2) + \dots + \pi(n-1)\pi(n) + \pi(n)\pi(1) = \sum_{k=1}^n k\pi'(k).$$

By the Cauchy-Schwarz inequality (cf. [N, p. 178]), for any $\pi \in S_n$ we have

$$\left(\sum_{k=1}^n k\pi(k) \right)^2 \leq \left(\sum_{k=1}^n k^2 \right) \left(\sum_{k=1}^n \pi(k)^2 \right)$$

and hence

$$\sum_{k=1}^n k\pi(k) \leq \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

If we let $\sigma(k) = n+1 - \pi(k)$ for all $k = 1, \dots, n$, then $\sigma \in S_n$ and

$$\begin{aligned} \sum_{k=1}^n k\pi(k) &= \sum_{k=1}^n k(n+1 - \sigma(k)) = (n+1) \sum_{k=1}^n k - \sum_{k=1}^n k\sigma(k) \\ &\geq \frac{n(n+1)^2}{2} - \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(n+2)}{6}. \end{aligned}$$

Thus

$$\left\{ \sum_{k=1}^n k\pi(k) : \pi \in S_n \right\} \subseteq T(n) := \left\{ \frac{n(n+1)(n+2)}{6}, \dots, \frac{n(n+1)(2n+1)}{6} \right\}. \quad (4.7)$$

Actually equality in (4.7) holds when $n \neq 3$, which was first realized by M. Aleksev (cf. the comments in [B]). Note that $|T(n)| = n(n^2 - 1)/6 + 1$.

Inspired by the above analysis, here we pose the following conjecture in additive combinatorics.

Conjecture 4.9. *Let $n \in \mathbb{Z}^+$ and let F be a field with $p(F) > n + 1$, where $p(F) = p$ if the characteristic of F is a prime p , and $p(F) = +\infty$ if the characteristic of F is zero. Let A be any finite subset of F with $|A| \geq n + \delta_{n,3}$, where $\delta_{n,3}$ is 1 or 0 according as $n = 3$ or not. Then, for the set*

$$S(A) := \left\{ \sum_{k=1}^n k a_k : a_1, \dots, a_n \text{ are distinct elements of } A \right\}, \quad (4.8)$$

we have

$$|S(A)| \geq \min \left\{ p(F), (|A| - n) \frac{n(n+1)}{2} + \frac{n(n^2-1)}{6} + 1 \right\}. \quad (4.9)$$

Remark 4.9. One may compare this conjecture with the author's conjectural linear extension of the Erdős-Heilbronn conjecture (cf. [SZ]). Perhaps, Conjecture 4.9 remains valid if we replace the field F by any finite additive group G with $|G| > 1$ and use $p(G)$ (the least prime factor of $|G|$) instead of $p(F)$.

Recall that the torsion subgroup of a group G is given by

$$\text{Tor}(G) = \{g \in G : g \text{ is of finite order}\}.$$

Conjecture 3.3(i) of the author [S19] states that if A is an n -subset (with $|A| = n > 2$) of an additive abelian group G of odd order then there is a numbering $a_1, \dots, a_n$ of all the elements of A such that $a_1 + a_2, \dots, a_{n-1} + a_n, a_n + a_1$ are pairwise distinct, this was verified by Yu-Xuan Ji (a student at Nanjing Univ.) for $|G| < 30$ in 2020. Motivated by this and Theorem 1.5, we formulate the following conjecture.

Conjecture 4.10. *Let G be an additive abelian group with $\text{Tor}(G)$ cyclic or $|\text{Tor}(G)|$ odd. For any finite subset A of G with $|A| = n > 3$, there is a numbering $a_1, \dots, a_n$ of all the elements of A such that the n sums*

$$a_1 + a_2 + a_3, a_2 + a_3 + a_4, \dots, a_{n-2} + a_{n-1} + a_n, a_{n-1} + a_n + a_1, a_n + a_1 + a_2$$

are pairwise distinct.

Remark 4.10. (a) Conjecture 4.10 holds in the case $A = G = \mathbb{Z}/n\mathbb{Z} = \{\bar{a} = a + n\mathbb{Z} : a \in \mathbb{Z}\}$ with $n > 3$ and $3 \nmid n$ since the natural list $\bar{0}, \bar{1}, \dots, \overline{n-1}$ of the elements of $\mathbb{Z}/n\mathbb{Z}$ meets the requirement.

(b) In 2008 the author [S08] proved that for any three n -subsets A, B, C of an additive abelian group G with $\text{Tor}(G)$ cyclic, there is a numbering $a_1, \dots, a_n$ of the elements of A , a numbering $b_1, \dots, b_n$ of the elements of B and a numbering $c_1, \dots, c_n$ of the elements of C such that the n sums $a_1 + b_1 + c_1, \dots, a_n + b_n + c_n$ are pairwise distinct.

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